Result 1 (chosen element is a member). [Lean, Classical.choose_spec]
Let A be a type. Let S be a subset of A. Assume (there exists $x : A,\ x \in S$) (`inhabited`). Then

$$\text{choose}(S, \mathit{inhabited}) \in S.$$

Result 2 (choice from an applied condition). [Lean, Classical.choose_spec]
Let A be a type. Let S be a subset of A. Assume (there exists $x : A,\ (y \mapsto y \in S)\,(x)$) (`inhabited`). Then

$$\text{choose}(S, \mathit{inhabited}) \in S.$$

Result 3 (choice with a universal condition). [Lean, Classical.choose_spec]
Let A be a type. Assume (there exists $x : A,$ (for every $y : A,\ y = x$)) (`inhabited`). Then

$$(\text{for every } y : A,\ y = \text{choose}(\mathit{inhabited}))\,.$$

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. Classical choice extracts witnesses noncomputably from existence evidence.